

0/1 Knapsack problem using Tabulation (Bottom-up Approach)

$n=4$, $W=8$, weights = { 3, 4, 5, 6 }
profits = { 2, 3, 4, 1 }

DP table

P_i	W_i	$i \backslash W$	0	1	2	3	4	5	6	7	8
		0	0	0	0	0	0	0	0	0	0
2	3	1	0	0	0	2	2	2	2	2	2
3	4	2	0	0	0	2	3	3	3	5	5
4	5	3	0	0	0	2	3	4	4	5	6
1	6	4	0	0	0	2	3	4	4	5	6

weight[3]=2
↑
previous Row
↑
 $8-5=3$
↑
 $\max(4+2, 5) = 6$
4 is current profit
2 is additional profit for remaining capacity

Procedure: Compare W_i with column weight for example in step 1

$3 > 0$ so we can not include item 1 so profit remains 0.

$3 > 2 \rightarrow 0$, $3 > 2$ include profit of 1st item & fill (2)

step 2: $4 > 1 \rightarrow 0$, $4 > 2 \rightarrow 0$, $4 > 3 \times$ include previous profit which is 2.

$4 > 4 \times$ now include current profit 3, then $4-4=0$ select $dp[i-1][0] \rightarrow 0$.
select $\max(3+0, 2) = 3$